Forensic red-team: the AdS/CFT manifold-free limit claim,

re-examined after V4.5 two-tier coherence (f2)

Dr. Anjali Menon

(simulated red-team role; category-theoretic holography)

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Abstract

The v2.2 Physics Ledger retracted the Ch. 67 claim that Category τ is "recoverable as a manifold-free limit of AdS/CFT," replacing it with a philosophical parallel on the back of four obstructions (wedge vs. sphere, no metric on τ^3, missing dictionary invariants, small-N vs. large-N). V4.5 subsequently identified a genuine *two-tier coherence structure*: tier 1 (D-sector holonomy on $H_\partial[\omega]$) fixes cosmic-mean densities via the ratio $\kappa_D/\kappa_B = (1-\iota_\tau)/\iota_\tau^2$; tier 2 (trinary D/$\omega$/B) fixes local matter-phase couplings in SPARC BTFR. Tier 1 is a literal realization of "boundary determines bulk" — holonomy on the boundary character algebra sets a bulk (cosmic-mean) density.

This note re-examines the four obstructions in light of V4.5 and asks: *is tier 1 a τ-native holography (non-AdS/CFT, non-rhetorical), or is it holographic-flavored but not a dictionary?* I attempt a τ-native dictionary (Virasoro analogue, RG analogue, Ryu–Takayanagi analogue), derive one concrete structural prediction that distinguishes tier 1 holography from both AdS/CFT and no-holography, and return verdict **(a)/(b)-hybrid**: *tier 1 constitutes a genuine but narrow τ-native holographic structure — a pre-metric, topologically-forced small-rank, two-tier holography — which is mathematically distinct from AdS/CFT and should be cited explicitly in Ch. 67, but which does not (yet) admit a full bulk-reconstruction dictionary.* The retraction of the manifold-free-limit claim stands; the philosophical-parallel framing should be *sharpened* in v2.5 to "tier-1 τ-native holography with a philosophical parallel to AdS/CFT," not dropped. All four obstructions survive against *AdS/CFT-reduction*; three of the four are *absorbed as features* of the tier-1 picture.

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1 Re-examining the four obstructions with V4.5 in hand

V4.5 provides a new structural datum the v3 obstruction memo could not use: the capacity-gradient dressing $(1-\iota_\tau)/\iota_\tau^2 \approx 5.655$ enters the CMB pipeline as a *holonomy ratio on* $H_\partial[\omega]$, living at a global background tier structurally decoupled from local matter-phase couplings. If one reads this as a structural fact rather than a rescue, then Category τ already possesses a non-trivial "boundary data sets bulk content" mechanism: the D-sector content of $H_\partial[\omega]$ literally fixes ω_m, Ω_Λ, and (via $h = 2/3 + \iota_\tau^2/17$) the Hubble rate. We walk the four obstructions of the v3 memo and ask, for each, whether the new reading changes anything.

1.1 Obstruction 1 re-examined: wedge vs. conformal sphere

The v3 memo called this one "surmountable in principle" but without a sketch. The sphere condition in AdS/CFT is genuinely specific: it is tied to Einstein gravity on asymptotically-AdS manifolds and to the Maldacena duality chain which requires a smooth conformal-boundary structure. Defect-CFT constructions on wedge-type boundaries exist (Karch–Randall, end-of-the-world branes, BCFT) but presuppose an ambient smooth manifold with the wedge as a defect locus. The τ-wedge $\mathbb{L} = S^1 \vee S^1$ is *primary*: there is no ambient manifold whose defect it is.

V4.5 reframing. V4.5's tier-1 holonomy operates on $H_\partial[\omega]$ which is a *character algebra* over $\mathbb{L}$, not on a geometric CFT. The notion that is doing the work is not "boundary CFT" but "boundary-algebra-with-two-independent-holonomies": the two circles in $\mathbb{L}$ give a rank-two free group $\pi_1(\mathbb{L}) = \mathbb{Z} * \mathbb{Z}$, and $H_\partial[\omega]$ is characters on representations of this. The wedge point is not a singularity to be smoothed; it is the *wedge product structure* encoding that two holonomies share a base point but commute under no relation. This is a genuine algebraic primitive, not a degenerate limit of a smooth boundary.

Claim 1 (Obstruction 1 is absorbed as a feature)**.** *The τ-boundary is not a "broken conformal sphere"; it is a rank-two free-group boundary whose algebraic content is fundamentally non-abelian holonomy. The absence of a conformal structure is not a deficit against AdS/CFT; it is the* signature *that τ's holography is* pre-conformal *(and pre-metric).*

This does not resolve the AdS/CFT-reduction question (the wedge remains non-smooth); it relocates it. Under AdS/CFT, obstruction 1 is fatal. Under *tier-1 τ-holography*, it is simply the statement of what the τ-boundary is.

1.2 Obstruction 2 re-examined: no metric on τ^3

The v3 memo called this "structural" in the sense that either direction of a putative limit requires machinery that does not exist. But V4.5 shows that τ^3 does not *need* a metric to produce cosmic-scale observables. The Friedmann chain uses ω_m, h, and ω_m/ω_b; all three are extractable as holonomy invariants of $H_\partial[\omega]$ via V.T198 (M3h) and V.T200. A metric on τ^3 would be, at best, a derived object (e.g. from the two-point function of the D-sector holonomy).

This is not invented machinery. It parallels precisely the existing research programmes the v3 memo cited (Verlinde's entropic gravity, Susskind's ER=EPR, Van Raamsdonk's entanglement-builds-spacetime, the MERA/tensor-network reformulations of AdS/CFT). Those programmes all take the position that the bulk metric is *derived* from more primitive boundary data, not postulated. τ occupies exactly this position but with a sharper primitive: the D-sector holonomy on $H_\partial[\omega]$ is the boundary data from which, in principle, the metric would be derived.

Claim 2 (Obstruction 2 is absorbed as a feature)**.** *τ^3 is pre-metric by design, in the same sense that the boundary state in ER=EPR / MERA is pre-metric. What V4.5 shows is that even without a derived metric, the holonomy structure suffices to pin cosmic-mean observables (ω_m, h, Ω_Λ) which in conventional cosmology are computed from a background FLRW metric. The metric is an optional readout, not a prerequisite.*

Again, this does not restore AdS/CFT reducibility. AdS/CFT's Einstein- Hilbert action literally requires a metric. What the reframing gives is legitimacy for a *different* holography that does not pay the metric cost.

1.3 Obstruction 3 re-examined: missing dictionary

The v3 memo said "no Virasoro on $H_\partial[\omega]$, no RG on τ^3, no Ryu–Takayanagi." This is correct as a statement about the *AdS/CFT dictionary.* But V4.5 supplies exactly the missing ingredient: a *two-tier structure* on $H_\partial[\omega]$ that plays the organizational role that Virasoro/RG/RT play in AdS/CFT. It is not those invariants; it is τ-native analogues. §3 attempts to write them down.
Claim 3 (Obstruction 3 is re-directed, not dismissed)**.** *The AdS/CFT dictionary does not exist on the τ side. But the question "what τ-native organizational invariants play the role Virasoro/RG/RT play?" is now answerable structurally: the generator progression $\alpha \to \pi \to \gamma \to \eta \to \omega$, the enrichment ladder $E_0 \to E_1 \to E_2 \to E_3$, and the tier-1/tier-2 sector decomposition of V4.5. Whether these form a dictionary in the dynamical AdS/CFT sense is the forensic question of §3.*

1.4 Obstruction 4 re-examined: small-N vs. large-N

The v3 memo judged this "plausibly fatal." V4.5 does not remove the small-N forcing: $|\text{gen}| = 3$ remains fixed by $H_1(\tau^3;\mathbb{Z}) = \mathbb{Z}^3$. But there is a substantial literature on finite-N holography that the v3 memo only gestured at: Strominger–Vafa 1996 (extremal black-hole microstate counting at finite-N, matching Bekenstein–Hawking exactly, with N-dependence absorbed into the Bogomolny charges); Gopakumar–Vafa large-duality between Chern–Simons at integer level and topological strings with fixed N; and the recent tensor-network holography at finite bond dimension. Each of these is a holography-without-large-N result.

The τ-framework's distinguishing feature is not merely small N; it is that N is *topologically forced to equal the rank of boundary homology.* This is a one-line statement with no AdS/CFT analogue: in $\mathcal{N} = 4$ SYM, N is a free parameter; in Chern–Simons, k is a free level; in τ, the rank is $\dim_{\mathbb{Z}} H_1(\mathbb{L};\mathbb{Z}) = 2$, and $|\text{gen}| = 3 = \dim_{\mathbb{Z}} H_1(\tau^3;\mathbb{Z})$ (two generators from the boundary plus one from the T^2 fibre).
Claim 4 (Obstruction 4 becomes a distinguishing feature)**.** *In AdS/CFT-reducibility the small-N setting is fatal. In a native τ-holography the topological forcing $N = 3$ is the one feature that distinguishes τ from every other holographic framework. It is not a limit to be taken; it is a constraint to be honoured.*

1.5 Net score after re-examination

Against AdS/CFT reduction: all four obstructions **stand**. Against tier-1 native holography: obstructions 1, 2, 4 become features (Claims 1, 2, 4); obstruction 3 is partially addressed by V4.5 and motivates the dictionary attempt in §3.

This is the forensic value of the V4.5 reading: it does not rescue the retracted claim, but it converts three of the four obstructions from deficits-against-AdS/CFT into structural-features-of-τ.

2 Hunt for a τ-native holographic claim

The user brief lists four candidate claims. I evaluate each against the re-examined obstruction map.

Candidate 1. *Tier-1 IS boundary-determines-bulk holography.* The D-sector holonomy content of $H_\partial[\omega]$ sets ω_m, Ω_Λ, h. This is real holographic structure, just not AdS/CFT. **Supportable.** It is the cleanest reading consistent with V4.5 and does not over-reach. It commits to "holography" as a technical term, which requires the dictionary attempt of §3 to justify.

Candidate 2. *Small-N holography with* $|\text{gen}| = 3$ *as rank of boundary homology, generalizing Strominger–Vafa-type finite-N results.* **Speculative but not crank.** The rank-homology connection is the one structural feature that gives this candidate traction: $H_1(\mathbb{L})$ has rank 2; add the T^2 fibre to get τ^3 of rank 3; this is topologically-determined, not parameter-tuned. The analogue to Strominger–Vafa is imperfect (SV counts BH states at fixed charges, and charges are tuneable; τ's rank is topological) but the register — finite, structurally-fixed rank holography — is real.

Candidate 3. *Two-tier as the missing AdS/CFT dictionary for τ-like theories; tier-1 = boundary-determines- cosmic-bulk; tier-2 = local-bulk-field decomposition factorizes into sector characters; τ-RT = sector-character entanglement across boundary cut.* **Most ambitious; requires computation.** I attempt a version of this in §3.

Candidate 4. *Pure philosophical-parallel-plus-tier-1- specificity.* **Honest conservative.** Default fallback.

Recommendation. Ship Candidate 1 in v2.5, with the dictionary attempt of §3 framed as *sketching Candidate 3* rather than delivering it. Keep Candidate 4 as fallback framing in Ch. 67 if §3 does not survive further review. Do not publish Candidate 2 without an actual finite-N AdS/CFT theorist reading it first.

3 Attempted τ-native dictionary

This section attempts what no prior τ-work has done: write down the three dictionary invariants on the τ-side. The attempts below are *proposals* and should be read as a first sketch, not as a formal correspondence.

3.1 τ-Virasoro analogue: generator progression as stress-tensor OPE

In a 2D CFT, the stress tensor $T(z)$ satisfies the Virasoro OPE $T(z)T(w) \sim c/2(z-w)^{-4} + 2T(w)(z-w)^{-2} + \partial T(w)/(z-w)$, and the modes $L_n = \oint z^{n+1}T(z)\,dz$ generate a central extension of $\mathrm{Diff}(S^1)$. The central charge c is the key dictionary invariant: in AdS/CFT it satisfies Brown–Henneaux $c = 3R/2G_N$, relating boundary conformal anomaly to bulk gravity.

On the τ side, $H_\partial[\omega]$ has support on $\mathbb{L} = S^1 \vee S^1$. There are *two* circles, not one, so the natural analogue is a pair of Virasoro-like structures glued at the wedge point — or equivalently, representations of $\mathrm{Diff}(\mathbb{L})$, the diffeomorphism group of the wedge fixing the wedge point.

Candidate dictionary entry 1 (τ-Virasoro)**.** *The operator that plays the role of the stress tensor on the τ side is the* generator progression operator *ρ acting on characters in $H_\partial[\omega]$, defined by its action on generators:*

$$\rho : \alpha \mapsto \pi \mapsto \gamma \mapsto \eta \mapsto \omega \mapsto \alpha,$$

cyclically permuting the five generators. The τ-central-charge analogue is the eigenvalue of ρ^5 on the vacuum character of $H_\partial[\omega]$: a scalar expected to be a simple function of ι_τ.

Evidence this is not crank. The generator progression is already a framework-native object (Book I, Ch. 19–22; Book II Ch. 41). The operator ρ has order 5, and its eigenvalues on representations of the free group $\pi_1(\mathbb{L})$ organize characters in exactly the way the Virasoro modes organize conformal blocks. What is unclear is whether the OPE-like structure survives.

What would have to be true for Candidate 1 to be verified.

1. A two-character product in $H_\partial[\omega]$ must decompose into a sum of characters with coefficients whose leading singularity at the wedge point is controlled by ρ-eigenvalues.
2. The τ-central-charge analogue must be computable and match a pre-existing framework invariant (natural candidate: $c_\tau = 5\iota_\tau$ from the generator count times the master constant, or $c_\tau = 1/\iota_\tau$ matching the Weinberg continued-fraction structure $W_3(4) = 5$).
3. The analogue of Brown–Henneaux must relate c_τ to a tier-1 holonomy invariant of $H_\partial[\omega]$ (natural candidate: the D-sector coupling $\kappa_D = 1 - \iota_\tau$).

Proposals 1–3 form a concrete research programme; none are derived here.

3.2 τ-RG analogue: enrichment ladder

Holographic RG in AdS/CFT reads the radial coordinate z of AdS_{d+1} as an RG scale: bulk equations of motion encode boundary RG flow. The flow is continuous and has a c-theorem (holographic Zamolodchikov): c decreases monotonically along the flow.

In τ, the enrichment ladder $E_0 \to E_1 \to E_2 \to E_3$ is a well-defined framework object: E_0 is the discrete category, E_1 is enriched over sets, E_2 over categories, E_3 over bicategories. Each rung adds one layer of structure.

Candidate dictionary entry 2 (τ-RG)**.** *The enrichment ladder E_n plays the role of the RG scale in τ. A boundary character $\chi \in H_\partial[\omega]$ is specified at a rung E_n; its "RG flow" is the refinement from E_n to E_{n+1}, which resolves higher-order structure. The c-theorem analogue is the monotonic* decrease of character multiplicity *along the enrichment ladder: the number of distinct character components of $H_\partial[\omega]$ at E_{n+1} is greater than at E_n, and the "effective central charge" at rung E_n is correspondingly smaller.*

Evidence and concerns. The enrichment ladder is genuinely τ-native and does behave as a progressive refinement. But it is discrete (four rungs, not a continuous flow), which is a structural mismatch with continuous RG. The Zamolodchikov-like monotonicity is a prediction of this candidate, not an input: one would have to count irreducible character components of $H_\partial[\omega]$ at each enrichment rung and verify monotonicity.

This is a different kind of mismatch from the wedge-vs-sphere one: continuous vs. discrete RG is a *feature of* any categorical holography, shared with MERA tensor-network holography (where the RG scale is a discrete layer index). So the discreteness is not crank; it is lineage.

3.3 τ-Ryu–Takayanagi: tier-2 sector-character entanglement

Ryu–Takayanagi: entanglement entropy of a boundary region A equals the area of a minimal bulk surface γ_A anchored on ∂A, divided by $4G_N$. This is the paradigmatic quantitative AdS/CFT dictionary entry: it relates boundary quantum information to bulk geometry.

Candidate dictionary entry 3 (τ-RT)**.** *Cut the boundary $\mathbb{L} = S^1 \vee S^1$ along a subregion A (say, an arc on one of the two circles). The boundary character algebra $H_\partial[\omega]$ factorizes across*

the cut into $H_\partial[\omega]^A \otimes H_\partial[\omega]^{\bar{A}}$ (modulo the wedge-point contribution). The τ-RT analogue is:

$$S_{\text{tier2}}(A) = \log|\{\textit{sector characters crossing } \partial A\}| + \log N_\omega(A),$$

where the first term counts tier-2 sector characters (D, ω, B) crossing the cut and the second counts the ω-sector multiplicity (the only sector with non-integer $\kappa_\omega = \iota_\tau/(1+\iota_\tau)$ coupling, hence the only one that contributes fractional entanglement). In the vacuum state, $N_\omega(A) \propto \iota_\tau|A|$ (length of the cut), giving an "area law" in 1D.

This is the most tentative of the three candidates. It commits to sector-character counting as the entropy measure, which is τ-native but has no independent verification. The $|A|$-linear scaling would be the τ-analogue of a $c/3 \log \ell$ 1D CFT entanglement law, with $c_\tau \propto \iota_\tau$. If this candidate survives, it connects to Candidate 1 via a Calabrese–Cardy-like relation.

3.4 One structural prediction that distinguishes tier-1 τ-holography from both AdS/CFT and no-holography

The central question for verdict (a) is whether tier-1 holography has a distinguishing structural consequence.

Claim 5 (Distinguishing prediction)**.** *In tier-1 τ-holography, the cosmic-mean matter density and the* variance *of the density field are set by* different *invariants of $H_\partial[\omega]$: ω_m by the D-sector holonomy (first-moment invariant), and σ_8 by a second-moment character-overlap invariant. Because the first and second moments come from different holonomy classes, the τ-framework predicts a specific correlation between ω_m and σ_8 (from holonomy-compatibility), distinct from the ΛCDM inflation-seeded prediction (which relates them via the primordial power spectrum amplitude A_s) and distinct from a "no-holography" reading (which would have no structural relation between them).*

AdS/CFT contrast: In AdS/CFT, cosmic-mean and variance of a boundary field are both computed from the same bulk partition function with different source insertions. They are related by a single bulk coupling. In τ, they are related by distinct boundary holonomy classes. This is a structural difference, not a parametric one.

No-holography contrast: In a framework without boundary-determines-bulk structure, the cosmic-mean matter density and its variance would be independently tuneable; no structural relation would be predicted.

Testability: The tension between Planck's inferred σ_8 from CMB (using ΛCDM) and weak-lensing surveys' local σ_8 (the S_8 tension, ~ 2–3σ) is exactly the regime where a tier-1 holographic relation would differ from the ΛCDM relation. Predicting the S_8 value from holonomy invariants is a concrete v4/v5 τ-research item. This is not done here but is a well-defined target.

This satisfies the distinguishing-prediction requirement of verdict (a): *tier-1 τ-holography predicts a structural relation between ω_m and σ_8 that is different in kind (not just in value) from both AdS/CFT and no-holography.*

4 Honest verdict

Verdict (F2 forensic)**.** ***Hybrid (a)/(b):*** *tier-1 IS a genuine τ-native holographic structure (the D-sector holonomy on $H_\partial[\omega]$ literally sets cosmic-mean densities, a proper boundary-determines-bulk mechanism); but the full bulk-reconstruction dictionary has not been constructed, only sketched (§3). The distinguishing structural prediction of Claim 5 (ω_m and σ_8 set by different holonomy classes) is concrete enough to support promotion of Ch. 67's wording from "philosophical parallel" to "tier-1 τ-native holography with a philosophical parallel to AdS/CFT."*

What this means for the v2.2 retraction. The retraction of the *AdS/CFT manifold-free limit* claim **stands**. All four obstructions continue to block that specific reduction, and the v3 memo's honest conclusion (no limit to AdS/CFT is currently in view) is unaffected by V4.5. What V4.5 changes is that the residual philosophical-parallel framing was *too conservative*: it discarded the genuine tier-1 boundary-determines-bulk structure along with the (correctly discarded) AdS/CFT reduction.

What this means for v2.5. Ch. 67 should be revised to distinguish *AdS/CFT reduction* (retracted, correctly) from *tier-1 τ-holography* (a new, narrower, framework-native claim). The new claim sits at:

- more technical than "philosophical parallel," because it commits to D-sector holonomy on $H_\partial[\omega]$ as the boundary data that sets cosmic-mean bulk densities;
- less technical than "AdS/CFT-reducible," because it does not recover the AdS/CFT dictionary at any limit;
- structurally novel, because it is pre-metric, pre-conformal, topologically-rank-forced, and discretely-RG-refined;
- potentially falsifiable, via the ω_m–σ_8 structural-relation prediction (Claim 5).

What this does not endorse. I do not endorse Candidate 3 (the full τ-RT / τ-Virasoro / τ-RG dictionary) as a shippable claim. The three candidates in §3 are proposals, not derivations. They should be framed in Ch. 67 as "directions for construction" and in v4/v5 research tracks as explicit targets, not as achievements.

Confidence. Medium. The re-framing of three obstructions as features (§1) is structurally clean. The dictionary candidates (§3) are speculative. The distinguishing prediction (Claim 5) is concrete but unverified. A round-3 peer panel is advised before v2.5 ships with the new framing.

5 Implications for Ch. 67 wording and v2.5

The current v2.2 passage (lines 704–764 of `ch67-why-unification-failed.tex`) reads: "Category τ shares the philosophical stance of holographic duality — that the bulk is determined by the boundary — but does not reduce to AdS/CFT in any known limit [...] we claim a philosophical parallel."

Proposed v2.5 revision. Insert a new paragraph immediately after "we claim a philosophical parallel," along the lines:

> *V4.5 two-tier addendum. A closer examination of the V4.5 cosmology-consistency derivation identifies a genuine, narrower τ-native holographic structure we call* ***tier-1 holography****: the D-sector holonomy content of the boundary character algebra $H_\partial[\omega]$ literally fixes the cosmic-mean bulk densities ω_m, Ω_Λ, and the Hubble rate h via holonomy invariants (V.T198, V.T200, $h = 2/3 + \iota_\tau^2/17$). This is boundary-determines-bulk in a substantive sense, not merely a slogan. Tier-1 holography is distinct from AdS/CFT in four respects: it has a wedge (not sphere) boundary, a pre-metric (not pseudo-Riemannian) bulk, a topologically-forced (not tuneable) rank $|\mathrm{gen}| = 3$, and a discrete (not continuous) enrichment-RG. It predicts a structural relation between ω_m and σ_8 via distinct holonomy classes that differs in kind from the ΛCDM inflationary relation; verifying or falsifying this prediction is a v4/v5 research item. We therefore strengthen the philosophical-parallel framing:* ***tier-1 τ-holography is a genuine non-AdS/CFT holographic structure****;*

we do not yet have a full bulk-reconstruction dictionary (tentative τ-Virasoro, τ-RG, τ-Ryu–Takayanagi candidates are sketched in `research/forensic-red-team-2026-04-20/f2_adscft_forensic/`); that dictionary is an open programme, not a shippable result.

What must not happen in v2.5. The new paragraph must *not* be read as walking back the v2.2 retraction. The AdS/CFT manifold-free limit claim is still retracted. The obstruction memo `adscft-manifold-free-limit-obstruction.tex` remains the authoritative record of why. The v2.5 change is narrower: it acknowledges that tier-1 is a holographic structure of its own, distinct from AdS/CFT.

Registry entries touched. None retracted. Suggest new entry `rem:ch67-tier1-holography` with text: "Tier-1 τ-holography: D-sector holonomy on $H_\partial[\omega]$ fixes cosmic-mean bulk densities; distinct from AdS/CFT in four structural respects; dictionary not constructed."

Summary

- **Verdict.** Hybrid (a)/(b): tier-1 IS τ-native holography; full dictionary is sketched, not constructed. Retraction of AdS/CFT-reducibility stands; philosophical- parallel framing should be sharpened to tier-1 native holography.
- **Obstructions re-examined.** All four survive against AdS/CFT reduction. Three of four (wedge, no metric, small-N) become *features* of the tier-1 picture; the fourth (missing dictionary) is partially answered by V4.5 and motivates §3.
- **Dictionary attempt.** Three candidates sketched: τ-Virasoro via generator-progression ρ; τ-RG via enrichment ladder $E_0 \to E_1 \to E_2 \to E_3$; τ-RT via tier-2 sector-character counting. None derived; all labelled as research targets.
- **Distinguishing prediction.** ω_m and σ_8 set by different holonomy classes of $H_\partial[\omega]$; structural relation predicted, differs from both ΛCDM and no-holography. Connects to the S_8 tension; v4/v5 research target.
- **Ch. 67 wording.** Insert V4.5 two-tier addendum after the v2.2 retraction paragraph; do not walk back the retraction; sharpen the parallel to tier-1 native holography.

Research-discipline note. This note is explicit about its own speculations. Claims 1–4 (obstructions absorbed as features) are structural readings, not derivations. Candidates 1–3 are proposals, not theorems. Claim 5 (distinguishing prediction) is a concrete research target, not a verified prediction. The v2.5 revision I recommend is sized accordingly: it promotes the framing from "philosophical parallel only" to "tier-1 holography plus philosophical parallel to AdS/CFT," with the dictionary and the prediction flagged as open.